Robot Control Systems 2025

Lab 2: DC Motor Modeling and Step-Response Lab

Purpose

Model the Qube-Servo 3 DC motor, derive its transfer functions, obtain the analytical step response, and validate the model via simulation and (optionally) on the hardware.

Learning Objectives

By the end of this lab you should be able to:

1) Write and interpret the electrical and mechanical equations of a DC motor.
2) Derive voltage→speed and voltage→position transfer functions.
3) Identify model coefficients (steady-state gain K and time constant τ) from physical parameters.
4) Compute the closed-form step responses and sketch them by hand.
5) Compare analytic predictions, transfer-function simulations, and device runs, and reason about discrepancies.

Equipment & Software

- Qube-Servo 3 (virtual and/or physical device).
- Python `3.11+` with `numpy`, `matplotlib`, `control`.
  Install: `pip install numpy matplotlib control`

1 Fundamental DC Motor Concepts

A DC motor converts electrical energy into mechanical energy through electromagnetic interaction. Consider the electromechanical system of a motor as shown in Figure 1. An RL series circuit can be used to model the internal resistance and inductance of the motor.

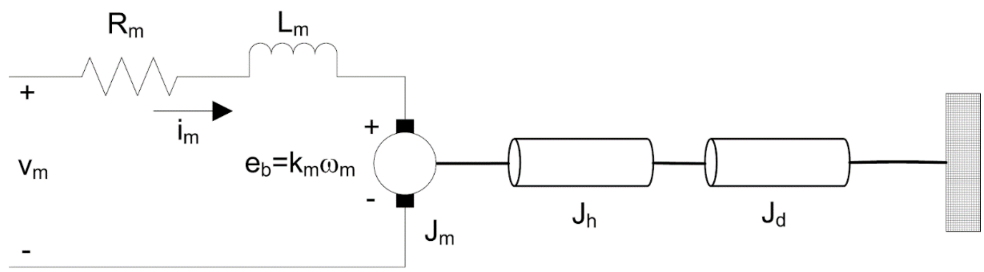

Figure 1: Typical Electromechanical Motor Configuration

a. Key Motor Parameters

i. Electrical Parameters

- Voltage (v_m): Applied electrical potential [V]
- Current (i_m): Current through motor windings [A]
- Resistance (R_m): Motor winding resistance [Ω]
- Inductance (L_m): Motor winding inductance [H]
- Back-EMF constant (k_m): Voltage generated per unit speed [V/(rad/s)]

ii. Mechanical Parameters

- Torque (τ_m): Motor output torque [N·m]
- Angular velocity (ω): Shaft rotational speed [rad/s]
- Inertia (J_e): Combined motor and load inertia [kg·m^2]
- Viscous friction (b): Damping coefficient [N·m·s/rad]
- Torque constant (k_t): Torque produced per unit current [N·m/A]

b. Basic Motor Equations

The equations from this section form the foundation for all subsequent modeling approaches discussed in this document.

i. Electrical Equations

In the circuit diagram in Figure 1, Kirchhoff's laws can dictate the relationship between the command voltage applied to the motor v_m as well as the voltage drops across the resistor v_r, inductor v_ℓ and motor armature e_b. The last term reflects the back-EMF generated by the spinning motor shaft and core.

$$v_m = v_r + v_\ell + e_b \tag{1}$$

The voltage drop across a resistor v_r with resistance R_m relates to the current flow i_m,

$$v_r = i_m R_m \tag{2}$$

The voltage drop across an inductor v_ℓ with inductance L_m relates to the rate of change of current,

$$v_\ell = L_m \frac{di_m}{dt} \tag{3}$$

The back-EMF (electromotive) voltage e_b is proportional to the speed of the motor shaft, ω_m, and the back-EMF constant of the motor, k_m.

$$e_b = k_m \, \omega_m \tag{4}$$

Substituting these expressions back into (1) yields the motor's electrical equation,

$$v_m = R_m i_m + L_m \frac{di_m}{dt} + k_m \omega_m \tag{5}$$

The inductance term is often dropped due to small effects on the overall model. The relevant electrical equation for a DC motor is therefore

$$v_m(t) = R_m i_m(t) + k_m \, \omega_m(t) \qquad \text{or} \qquad v_m(t) = R_m i_m(t) + k_m \, \dot{\theta}_m(t) \tag{6}$$

where $v_m(t)$ is the applied voltage (control input), R_m is the motor's resistance, $i_m(t)$ is the motor current, k_m is the back-EMF constant, and $\omega_m(t) = \dot{\theta}_m(t)$ is the angular velocity ($\theta_m(t)$ is the angular position of the motor shaft).

ii. Mechanical Equations

We start with Newton's law applied to rotational systems. The net motor torque τ_{net} is

$$\tau_{\text{net}} = J_{\text{eq}} \, \dot{\omega} \tag{7}$$

where J_{eq} is the total moment of inertia of the motor shaft, hub and associated load in $\mathrm{kg\,m^2}$ and $\dot{\omega}$ is the angular acceleration of the motor.

The equivalent moment of inertia can be calculated as the sum of the individual components,

$$J_{\text{eq}} = J_m + J_h + J_d \tag{8}$$

where J_m relates to the mass moment of inertia of the motor, J_h may represent a hub/attachment, and J_d represents the moment of inertia of the attached load.

The moment of inertia of a disk about its pivot, with mass m and radius r, is

$$J_d = \frac{1}{2} m r^2 \,. \tag{9}$$

The net torque on the motor is given by

$$\tau_{\text{net}} = \tau_{\text{app}} - b\,\omega \tag{10}$$

where τ_{app} is the applied motor torque, ω is the motor speed, and b represents motor damping. Combining with the inertia relation yields the applied torque equation

$$\tau_{\text{app}} = J_{\text{eq}}\,\dot{\omega} + b\,\omega \,. \tag{11}$$

If we use $\theta_m(t)$ as the angular position and $\omega = \dot{\theta}_m$, the above becomes

$$\tau_{\text{app}} = J_{\text{eq}}\,\ddot{\theta}_m + b\,\dot{\theta}_m \,. \tag{12}$$

Finally, the applied torque is directly proportional to the motor current generated as a result of the applied voltage command v_m:

$$\tau_{\text{app}} = k_t\, i_m \,, \tag{13}$$

where k_t is the motor torque constant in $\mathrm{N\,m/A}$.

2 Qube-Servo 3 Parameters

Use the following nominal values for modeling and numerical substitution.

Table 1: Qube-Servo 3 key parameters

Symbol	**Description**	**Value**
V_{nom}	Nominal input voltage	24 V
τ_{nom}	Nominal torque	20.4 mN m
ω_{nom}	Nominal speed	5400 RPM
I_{nom}	Nominal current	0.5 A
R_m	Terminal resistance	7.5 Ω
k_t	Torque constant	0.0422 N m/A
k_m	Back-EMF constant	0.0422 V/(rad/s)
J_m	Rotor inertia	1.4×10^{-6} kg m^2
L_m	Rotor inductance	1.15 mH
m_h	Module hub mass	0.0106 kg
r_h	Module hub radius	0.0111 m
J_h	Hub moment of inertia	0.6×10^{-6} kg m^2
m_d	Disc mass	0.053 kg
r_d	Disc radius	0.0248 m
m_r	Rotary arm mass	0.095 kg
L_r	Rotary arm length (pivot→end)	0.085 m
m_p	Pendulum link mass	0.024 kg
L_p	Pendulum link length	0.129 m
Encoder lines (x1 / x4)		512 / 2048 lines/rev
Encoder resolution		0.176 deg/count, 0.003 07 rad/count
Amplifier type		PWM
Peak stall current		1.8 A@15 V
Continuous current		0.58 A@5 V
Output voltage (rec./max)		±10 V / ±15 V
Default deadband		0.65 V

3 Student Tasks

3.1 Task 1 — Derive the Transfer Functions

Starting from (6)–(13):

1. Eliminate i_m using (6) and (13).

2. Derive the voltage-to-speed transfer function $\dfrac{\Omega_m(s)}{V_m(s)}$.

3. Derive the voltage-to-position transfer function $\dfrac{\Theta_m(s)}{V_m(s)}$ by introducing the integrator between ω_m and θ_m.

4. Standard form of the transfer function for a first-order system is

$$G(s) = \frac{K}{\tau s + 1}.$$

Express each transfer function in standard form and **identify all coefficients** (steady-state gain K, time constant τ).

3.2 Task 2 — Substitute Qube-Servo 3 Coefficients

Using Table 1:

1. Compute J_{eq}.

2. Provide numeric K and τ for the chosen model.

3.3 Task 3 — Analytical Step Response

For a unit-step signal applied to your *speed* model:

1. Use inverse Laplace to find $\omega_m(t)$ in closed form (for zero initial conditions).

2. For the *position* model, integrate accordingly or perform inverse Laplace on $\Theta_m(s)$ directly.

3. **Plot by hand** the expected step response(s). Mark $t = \tau$ and the 63.2% point.

3.4 Task 4 — Simulation Study

Use the provided `Lab2.ipynb` as a code skeleton and ensure `pip install control`. Run three **distinct** simulations with the same input (state amplitude and step time):

a) **Transfer function**: simulate $\Omega_m(s)/V_m(s)$ and/or $\Theta_m(s)/V_m(s)$ in Python and plot.

b) **Analytic ODE**: evaluate your closed forms $\omega_m(t)$ and $\theta_m(t)$ over the same time grid and compare to (a).

c) **QServo (virtual → hardware)**: apply the same step to the Qube-Servo 3 (first the virtual device, then physical). Record θ [rad] and ω [rad/s] (e.g., `MotorPosition`, `MotorSpeed`). Discuss discrepancies (deadband, friction b, supply limits, PWM, encoder quantization, parameter mismatch).